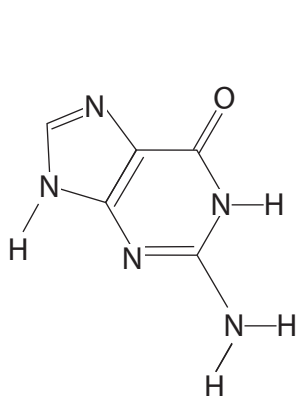


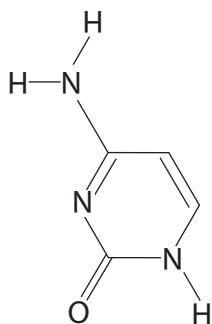
- 8 Mononucleotides are joined together to form DNA. Each mononucleotide is composed of a deoxyribose linked to a phosphate and one of four bases.

(a) The diagrams show the structure of the four bases.

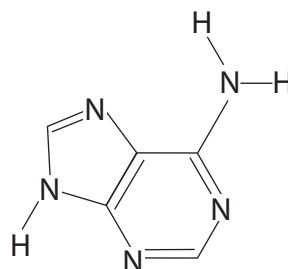
Guanine and adenine have two ring structures, cytosine and thymine have one ring structure.



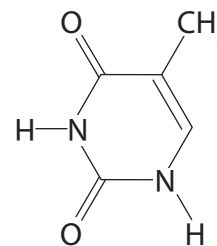
guanine



cytosine



adenine



thymine

Explain the role of these bases in the structure of DNA.

Use the information in the diagram to support your answer.

(3)

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DO NOT WRITE IN THIS AREA

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- (b) The DNA of the bacterium *E. coli* has a mass of 3.1×10^9 daltons (Da) and consists of 4.7×10^6 base pairs.

- (i) Calculate the mean mass of a base pair.

Express your answer in standard form.

(1)

Answer Da

- (ii) Calculate the total mass of DNA produced after three cell divisions.

Give your answer in grams, where 1 dalton = 1.67×10^{-24} grams.

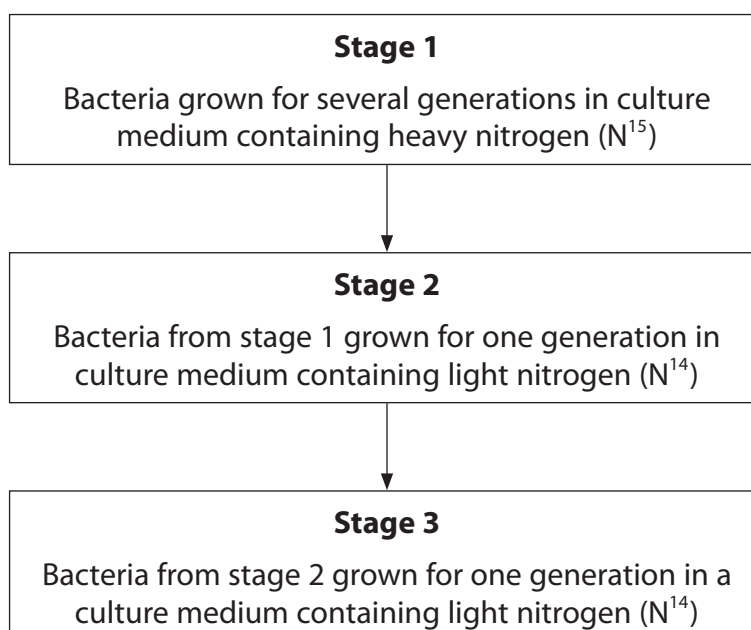
(2)

Answer g

- (c) Meselson and Stahl provided data that supported the theory that DNA replicates by semi-conservative replication and disproved competing theories.

Heavy nitrogen (N^{15}) and light nitrogen (N^{14}) were used in their experiments.

The flow chart summarises part of one experiment performed by Meselson and Stahl.



After each stage, a sample of DNA was taken from the bacteria and the DNA molecules separated on a density gradient in a tube.

The heavier DNA molecules form bands lower down the gradient than the lighter DNA molecules.

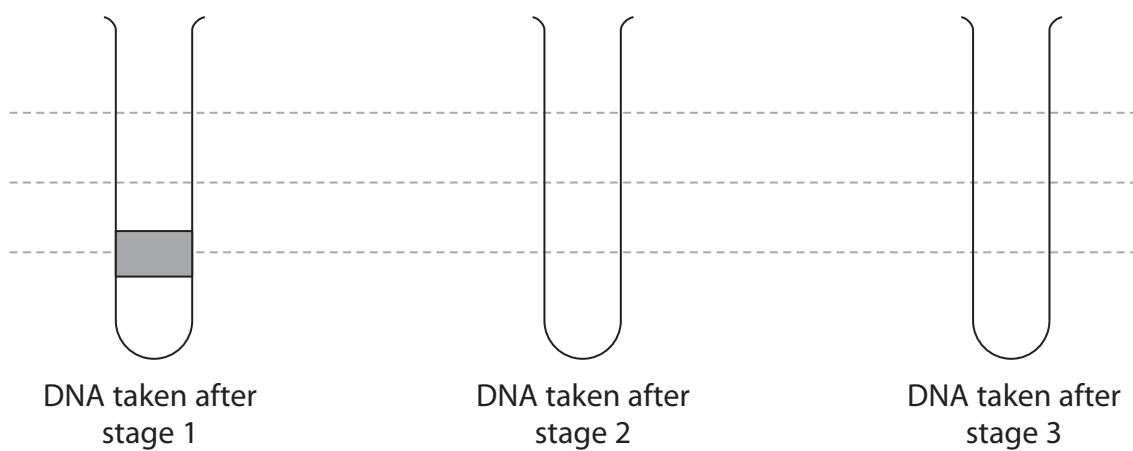
The width of each band is proportional to the percentage of molecules in the sample.

- (i) Complete the diagram to show the results of this investigation.

Use the dotted lines to help you position the bands on the diagram.

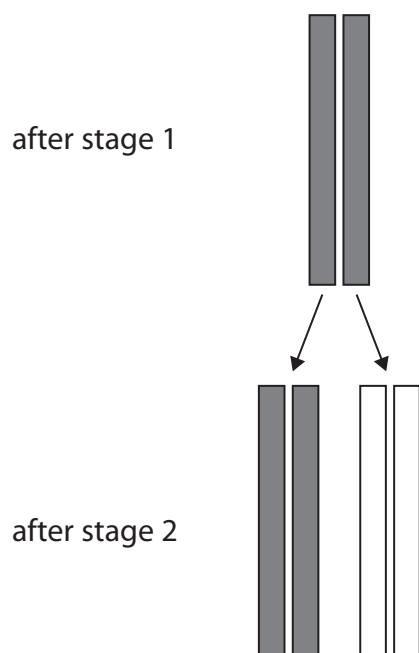
The first one has been done for you.

(3)



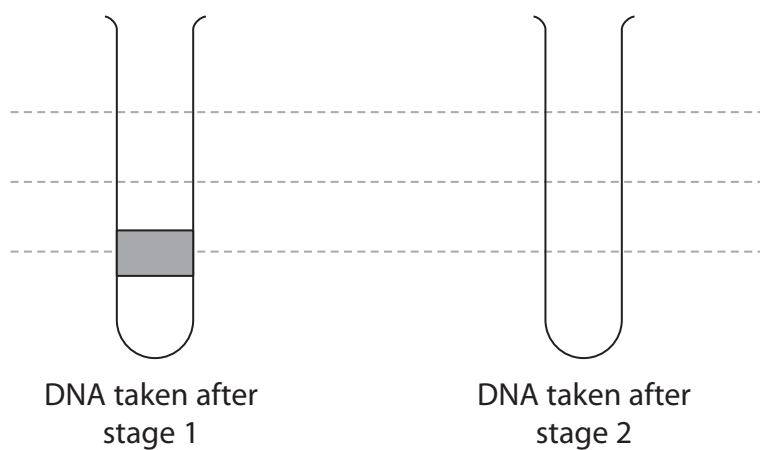
(ii) One competing theory was the conservative theory.

The diagram shows the conservative theory for the replication of DNA.



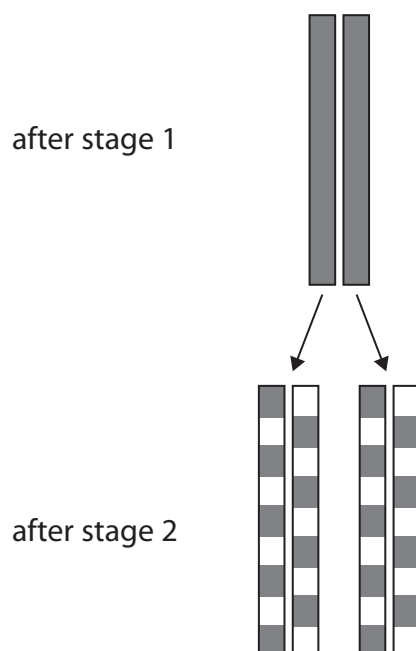
Complete the diagram to show the position of the bands on the density gradient in a tube if this was the correct theory.

(2)



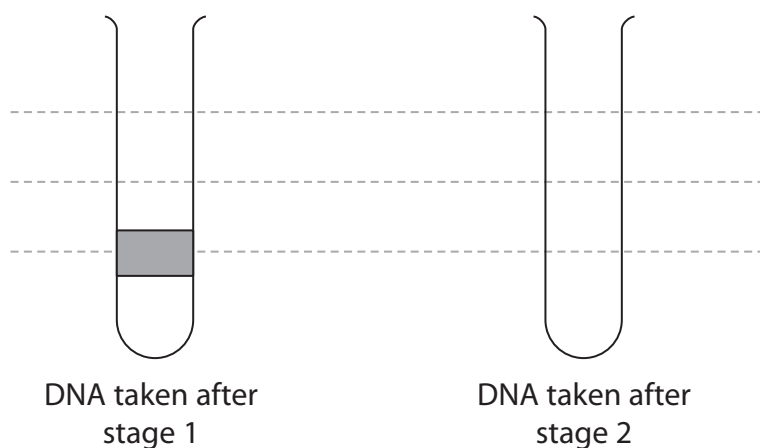
(iii) Another competing theory was the dispersive theory.

The diagram shows the dispersive theory.



Complete the diagram to show the position of the bands on the density gradient in a tube if this was the correct theory.

(2)



(Total for Question 8 = 13 marks)

TOTAL FOR PAPER = 80 MARKS

